

From Representation Credit to Cellular Credit: Testing pART-Inspired Models of Dendritic Credit Assignment

Ognjen Čadovski

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ognjen.cadovski@robbit.rs

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Abstract

A delayed behavioral outcome can reinforce an action, but it does not directly specify how individual neurons with opposing causal roles should change. We examined this distinction in a sequence of abstract closed-loop brain–computer interface simulations motivated by role-dependent dendritic signals reported by Francioni et al. [Francioni et al., 2026]. The studies progressively tested selective routing of an existing opponent pattern, scalar reinforcement of hypotheses drawn from a random repertoire, learned top-down expectations, and a local SMART-inspired plasticity motif. Scalar reinforcement reliably selected useful distributed representations, maintained them across delay, and supported remapping. The integrated theory-led models, however, did not satisfy a preregistered longitudinal criterion linking an early dendritic residual to later synaptic and somatic change. Further analysis showed that learned top-down topology was almost exactly inherited from outcome-selected activity in a fixed repertoire. A separate, explicitly non-Grossberg comparator combined common scalar advantage with cell-local exploratory eligibility. It learned arbitrary hidden sign maps and relearned them after remapping; at population size $N = 32$, alignment was 0.987 after acquisition and 0.964 after remapping. These results distinguish two computational claims. Scalar outcome is sufficient for representation-level credit in the tested pART-inspired models, and it is sufficient for arbitrary cellular credit when an additional local eligibility trace is supplied. The simulations do not establish that this eligibility rule is implemented biologically or contained in published Grossberg theory.

1 Introduction

Neural learning faces two related credit-assignment problems. A consequence may arrive after the action that caused it, creating a temporal problem, and simultaneously active neurons may contribute differently to that consequence, creating a structural problem. The second difficulty is especially clear in brain–computer interface (BCI) tasks in which intermingled cortical cells have opposite causal roles. Francioni et al. reported role-dependent dendritic signals during such learning and interpreted them as vectorized instructive signals [Francioni et al., 2026]. One possible mechanism would compute a neuron-indexed teaching vector. Another would broadcast only a scalar outcome and allow local state or recent history to determine the sign of each update.

pART, ART, and SMART offer components that make the second possibility worth testing. They provide mechanisms for category selection, maintenance across delay, learned expectations, matched feedback, and local plasticity. None of those components alone resolves the key information question: where does an arbitrary cell-specific sign pattern originate when the environment returns only a scalar result?

We addressed that question through a cumulative computational program. Each study removed a shortcut or supplied a mechanism left unresolved by the preceding one. Selective routing was tested first as an existence proof. Hidden cellular identity was then removed by introducing a random repertoire; learned top-down expectations replaced copied patterns; a reduced local plasticity motif was tested before integration; and the origin of the learned topology was finally isolated. When the theory-led sequence did not construct arbitrary cellular topology, a source audit was followed by a generic local-eligibility diagnostic. This order matters because a successful behavior does not by itself identify the level at which credit was assigned.

The contribution is therefore both computational and methodological. The experiments separate representation selection from cellular sign learning, evaluate a prospective dendrite-to-synapse-to-soma criterion, and identify the additional information that made the generic cellular solution possible. All numerical results and figures come from the archived repository record; claims about published theory and biology are supported separately by external citations.

2 Related work

ART addresses stability–plasticity through category matching, vigilance, reset/search, and learned top-down expectations [Carpenter and Grossberg, 1987]. pART emphasizes structural and temporal credit: a representation can be selected, maintained over a delay, and reinforced by broad motivational signals [Grossberg, 2018]. These mechanisms motivate representation-level credit but do not, by themselves, specify an arbitrary neuron-resolved feedback vector.

SMART supplies a more cellular circuit motif. In a visual thalamocortical setting, learned higher-to-lower feedback reaches apical dendrites, a match-dependent on-center/off-surround circuit changes timing and participation, and a local voltage/conductance-gated rule changes synapses [Grossberg and Versace, 2008]. Canonical laminar extensions describe related modulatory feedback motifs [Grossberg, 2021]. Grossberg’s vector associative motor systems address a different information boundary: they use an available target-versus-present difference vector [Gaudiano and Grossberg, 1991]. Importing such a vector into the present task would assume the neuron-specific information that the simulations are intended to explain.

Three computational accounts must consequently be distinguished. A centralized system may compute and route a cell-wise error. A representation-level system may reinforce one useful distributed pattern already present in a repertoire. A local exploratory system may instead multiply a common scalar advantage by a cell-specific eligibility trace. The experiments below test the latter two possibilities without treating them as equivalent.

3 Methods

3.1 Task and information boundary

The experiments use abstract closed-loop BCI tasks with hidden, balanced causal roles $c_i \in \{-1, +1\}$ assigned to intermingled cells. The learner observes context or state, its own activity, permitted local variables, and scalar outcome. The role vector c remains environment-only except in explicitly labeled positive controls and offline analyses. Primary endpoints include task performance, correlation or alignment of learned patterns with c , recovery after an independently changed hidden map, and changes under interventions that remove proposed causal ingredients.

The theory-led models distinguish a selected category–hypothesis representation $H_{k,h}$, a learned expectation $T_{k,h,i}$, a modeled top-down or apical quantity D_i , plastic lower weights W_i , and somatic

output S_i . Their prospective longitudinal test is

$$D_{\text{residual},i}^{\text{early}} \longrightarrow \Delta W_i \longrightarrow \Delta S_i, \quad (1)$$

evaluated within a held hypothesis so that switching between patterns cannot create an apparent effect.

SMART-derived models implement the qualitative local dependency

$$\Delta W_{jk} \propto f_G(V_k, \bar{g}_{jk}) [\bar{g}_{jk} f_N(V_k)(\hat{w} - \check{w}) + w_0 - w_{jk}], \quad (2)$$

where conductance, voltage, and spike timing determine local change [Grossberg and Versace, 2008]. The generic local-eligibility comparator uses a different update,

$$\Delta A_i = \eta_N(R - b)e_i, \quad (3)$$

where $R - b$ is a common scalar reward advantage, e_i is a retained local exploratory perturbation, and η_N corrects known population-mean scaling. Hidden roles do not enter this update.

3.2 Experimental sequence and reproducibility

Later studies used frozen held-out seed sets, preregistered effect floors, seed-level bootstrap intervals, explicit positive controls supplied with hidden vector information, and causal ablations such as shuffled outcome, removed exploration, or removed eligibility. Table 1 provides the only manuscript-level crosswalk to the repository’s internal identifiers. The Results retain this chronology but use scientific study names in the narrative.

3.3 Claim boundaries

Quantitative statements labeled as results are derived from committed protocols, frozen analyses, held-out records, or explicitly identified post-hoc audits in the project repository [OgnjenX, 2026]. No new simulations were conducted for this manuscript. The prestructured study tests routing, not de novo sign learning. Conclusions from theory-led studies apply to the implemented compositions and endpoints rather than to all ART, pART, or SMART models. The local-eligibility comparator is a generic three-factor rule and is not attributed to Grossberg. Biological interpretations remain hypotheses unless supported by the cited experimental literature.

3.4 Theoretical bridge audit

A source audit mapped representation selection and maintenance to pART, expectation learning and match/reset to ART, apical feedback and timing-dependent local plasticity to SMART, and the dendritic observation statistic to Francioni et al. The weakest bridge was acquisition of arbitrary neuron-fine $H \rightarrow T$ topology from scalar outcome. Two additional assumptions remained outside the cited theories: mapping the SMART surround to NDNF-sensitive dendritic signals, and mapping modeled current or voltage to a soma-conditioned calcium residual. These gaps determined the later topology and local-eligibility tests.

Registry ID	Study	Evidential role
EXP000	Prestructured opponent-code routing	Existence proof for routing an already available pattern.
EXP001	Random-repertoire selection	Scalar selection and delayed credit at the representation level.
EXP002	Learned top-down expectancy	Test of whether a learned expectation supplies the longitudinal cellular signal.
EXP003– EXP003b	SMART motif and integrated composition	Local mechanism validation followed by a prospective integrated test.
EXP004	Origin of learned topology	Test of construction versus inheritance from fixed repertoire geometry.
EXP005	Source audit and local eligibility	Source-boundary audit followed by a generic cellular-credit diagnostic.

Table 1: Crosswalk between repository identifiers and the scientific studies described in the manuscript.

4 Results

4.1 Prestructured opponent-code routing

The first study corresponds to registry record EXP000. After two retained calibration attempts, the corrected model achieved late accuracy 0.978 across 30 seeds, with P+ modulation +0.673, P- modulation −0.673, and an opposition index of +1.346 (Figure 1). Selective feedback could therefore express a strong opponent pattern without receiving an explicit vector error on each trial.

The same result exposed the study’s structural shortcut. P+/P- identity was recoverable from even and odd neuron indices and was already encoded in category templates. Freezing category and value learning left late accuracy unchanged at 0.978, while ordinary corrected trials showed 100% resonance and no resets. The result established routing of an existing pattern, but it left open how a useful pattern could be found when cellular identity was hidden. That question motivated the random-repertoire study.

4.2 Random-repertoire selection

The next study (EXP001) removed visible population identity and replaced the opponent template with a random hypothesis bank. Scalar outcome selected useful distributed hypotheses, maintained them across irrelevant events, and supported relearning after the hidden roles changed. The confirmatory comparison also showed that post-remap success depended strongly on the available repertoire and that the best initial alignment increased with bank capacity (Figure 2).

This capacity dependence located the achievement at the representation level. A useful direction had to occur in the bank before reinforcement could select it. Instantaneous dendritic-residual alignment also appeared in non-learning controls and did not reliably predict later cell-specific activity change, whereas a neuron-indexed positive control did. The model could learn which available pattern worked, but not independently infer the sign required at each cell. A learned expectation was introduced next to test whether repeated outcome-selected activity could generate that missing signal.

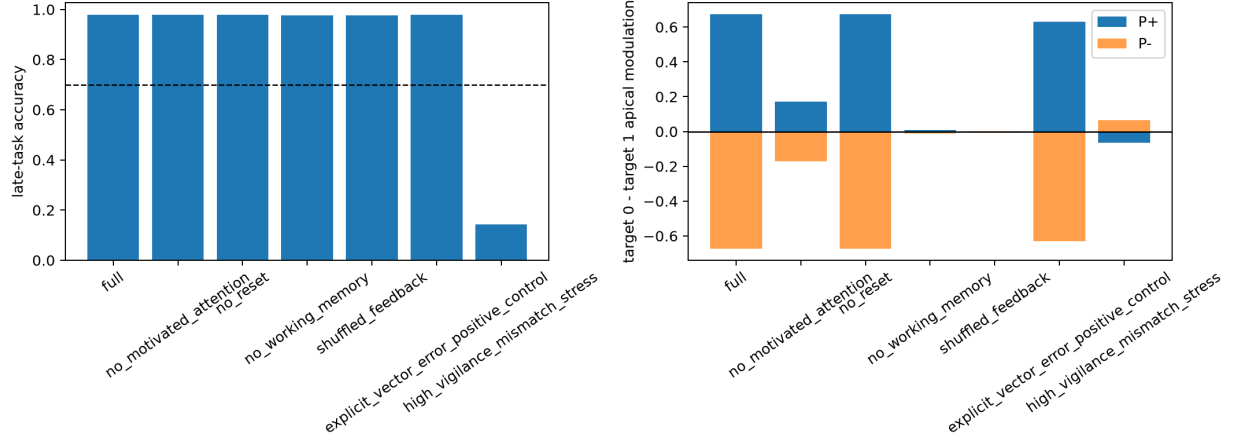


Figure 1: Repository-derived summary of prestructured routing. Late-task accuracy remains high under several ablations because the opponent pattern is already present in the model; shuffled feedback disrupts P+/P- modulation, while the explicit vector control serves as a positive control.

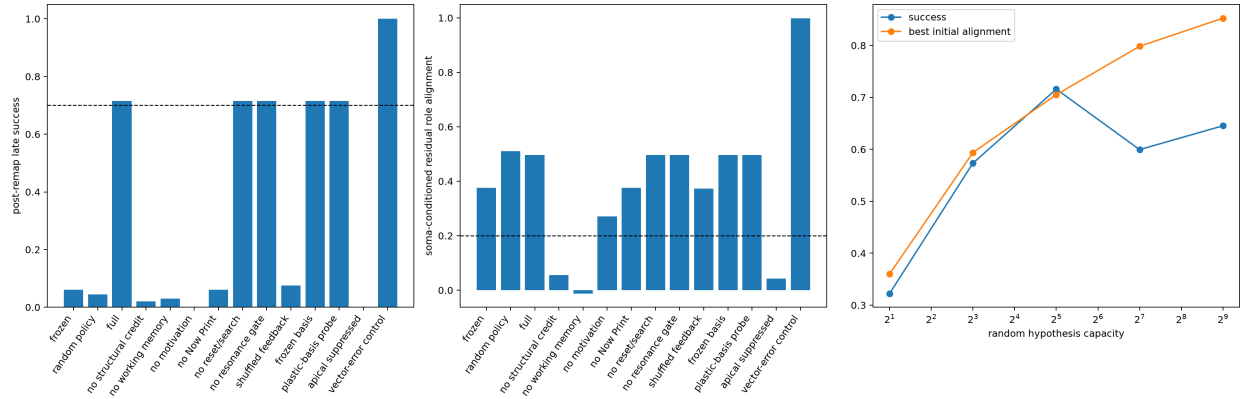


Figure 2: Frozen random-repertoire comparison. Left: post-remap success by condition. Center: soma-conditioned residual alignment. Right: success and the best initial role alignment as hypothesis-bank capacity increases. The capacity relation is consistent with selection from a fixed repertoire rather than de novo cellular sign discovery.

4.3 Learned top-down expectancy

The learned-expectancy study (EXP002) added a Grossberg outstar-derived association,

$$T_h \leftarrow T_h + \eta(S_{\sim} - T_h), \quad (4)$$

to store somatic patterns repeatedly coactive with selected hypotheses. On frozen data, the primary model reached behavioral scores of 0.528 before and 0.397 after remapping, while top-down role alignment reached 0.343 and 0.287. The expectation was therefore learned and remappable.

Behavior alone did not distinguish this mechanism from simpler alternatives. Directly copying the chosen pattern produced similar alignment, and a contextual bandit matched behavior. More importantly, the pre- and post-remap longitudinal residual effects were -0.326 and -0.016 , compared with 0.660 and 0.545 for the neuron-indexed control. Suppressing expectation learning removed learned pattern alignment without materially changing behavior. These results shifted the question from whether an expectation could be stored to whether supplied feedback could drive a local plasticity mechanism and later alter somatic output.

4.4 From a SMART motif to an integrated composition

The local-mechanism program began with a source audit (EXP003) and a reduced SMART-inspired motif (EXP003a). Supplied matched feedback altered spike timing, gated local plasticity, and changed a later feedforward response. The modeled update increased from 0.600 to 0.731 , and a learning-off probe changed from 0 to 7 of 8 spikes. Competitor, top-down-ablated, and mismatch conditions produced no update. This isolated the local arrow from matched feedback to plasticity without yet solving behavioral credit assignment.

The integrated composition (EXP003b) then combined the motif with delayed structural credit and learned expectations. Across 12 held-out mappings, feedback advanced local spike timing by 6.29 ms and changed lower weights. The causal contrast disappeared when SMART input, learned expectation, or local learning was removed. Nevertheless, the prospective longitudinal endpoint was not met: early soma-conditioned dendritic residuals did not predict lower-weight or held-hypothesis somatic change at the frozen threshold, while the neuron-wise positive control completed the chain.

An explicitly post-hoc failure localization found that the learned expectation T and quantities closer to the plasticity rule predicted sparse ΔW , whereas the soma-conditioned residual discarded much of that information. Raw $T \rightarrow \Delta W$ correlation was 0.416 [$0.290, 0.530$], but residual $D \rightarrow \Delta W$ correlation was -0.004 [$-0.141, 0.128$]. The integrated result therefore redirected the program toward the origin of the expectation topology and the adequacy of the observation model.

4.5 Origin of neuron-specific topology

The topology-origin study (EXP004) prospectively varied the best role coverage available in a fixed bank. At bank size $M = 16$, low, medium, and high initial coverage produced behavioral scores of 0.765 , 0.882 , and 0.962 and learned- T alignments of 0.258 , 0.327 , and 0.395 (Figure 3). The high-minus-low behavioral difference was 0.197 [$0.137, 0.255$]. Across random banks, candidate coverage increased with M , although larger banks also made search more difficult.

A constructed sequential condition demonstrated behavioral composition: performance reached 0.980 and exceeded the best repeated-single action by 0.313 [$0.260, 0.374$]. A contextual bandit, however, reached 0.977 . More decisively, the learned expectation was almost identical to experienced activity:

$$\text{corr}(T, \text{selected initial soma}) = 0.999851, \quad \text{corr}(T, \text{mean target}) = 0.999870. \quad (5)$$

Exact replay reconstructed T with RMSE 0.0, and T did not surpass the best individual experienced target. Outcome permutation altered topology mainly by changing which hypotheses and targets were visited. Thus the implemented outstar consolidated outcome-selected states, while their neuron coordinates came primarily from fixed repertoire geometry. This finding made the missing operation precise: arbitrary cellular topology required a cell-varying quantity that was not already embedded in a candidate pattern.

4.6 Source audit and a local-eligibility diagnostic

The final source audit (EXP005) asked whether a published Grossberg equation supplied the missing operation. pART, ART/outstar learning, exploration, SMART, and vector motor systems each supplied adjacent functions, but the audit found no source-derived rule that joined an independent local exploratory trace to delayed scalar consequence in order to learn arbitrary intermingled signs. A new Grossberg primary model was therefore not constructed.

A separately labeled generic comparator tested the computational sufficiency of that additional operation. On 16 held-out maps, after 1,280 acquisition episodes and 2,560 remap episodes, alignment was 0.997, 0.993, 0.987, and 0.972 for $N = 8, 16, 32, 64$; post-remap alignment was 0.968, 0.960, 0.964, and 0.950. At $N = 32$, the learning trajectory rose rapidly during acquisition and recovered after an independent hidden remap (Figure 4). Shuffling outcome reduced post-remap alignment to 0.229. Removing plasticity, temporal eligibility, or exploration left alignment near zero, and generic-minus-shuffled alignment was 0.735 [0.655, 0.820]. Changed-cell sign reversal was perfect through $N = 32$ and 0.975 at $N = 64$.

Exact replay reconstructed the updates to numerical precision, confirming that no hidden role information entered the learner. Final topology also exceeded the best single exploratory sample at every population size, showing accumulation rather than selection of one lucky perturbation. The decisive difference from the preceding models was not a richer scalar reward. It was the private eligibility e_i , which supplied the cell-varying factor needed to turn a common scalar advantage into different local updates.

5 Discussion

The sequence separates two levels of credit that can otherwise look similar in successful behavior. Scalar reinforcement was sufficient to select and maintain useful distributed representations. It also supported learned expectations and remapping. Yet the neuron-specific structure expressed by those models was largely inherited from the repertoire of patterns that could be selected. Pattern selection answers “which available representation worked?”; it does not necessarily answer “which individual cells should change sign?”

The generic comparator supplied a concrete answer to the second question. Every cell received the same advantage signal, but each combined it with a private memory of its own recent perturbation. Repeated trials accumulated those locally signed correlations into an arbitrary hidden topology. This demonstrates that a centrally routed vector error is not mathematically necessary for the abstract task. It does not determine whether biological learning uses perturbation eligibility, explicit routed instruction, matched expectation, or a mixture of these mechanisms.

The integrated-model results also expose a measurement problem. Information present in a learned expectation, current, or spike-timing variable need not survive conversion to a sparse soma-conditioned calcium residual. The post-hoc localization is useful because it identifies where information was lost, but it cannot retroactively satisfy the failed prospective endpoint. A future

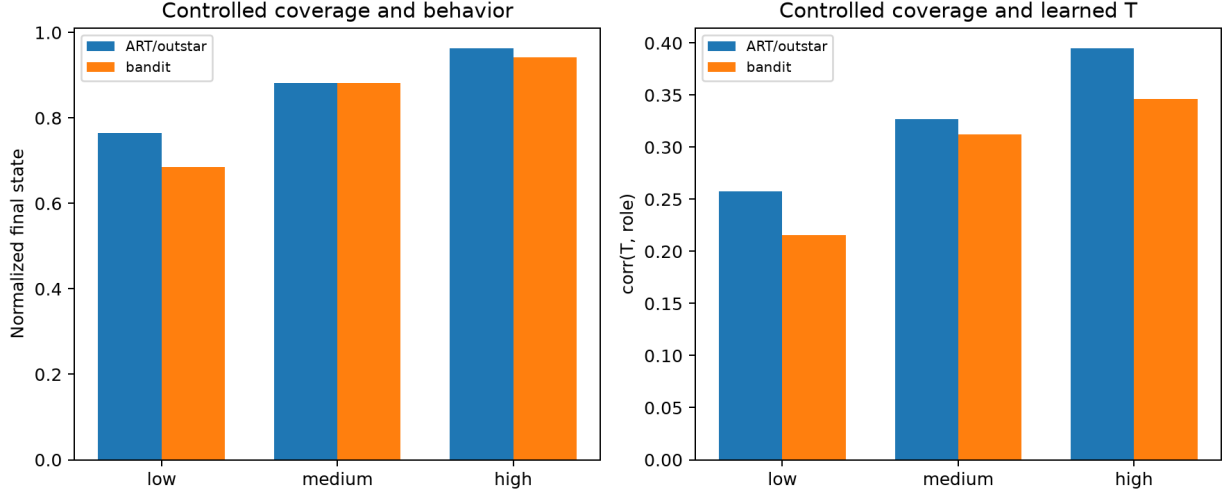


Figure 3: Frozen controlled-coverage result. Both final behavior (left) and alignment of the learned top-down expectation T with hidden roles (right) increase with the quality of topology already available in the bank. The contextual-bandit comparison follows the same trend.

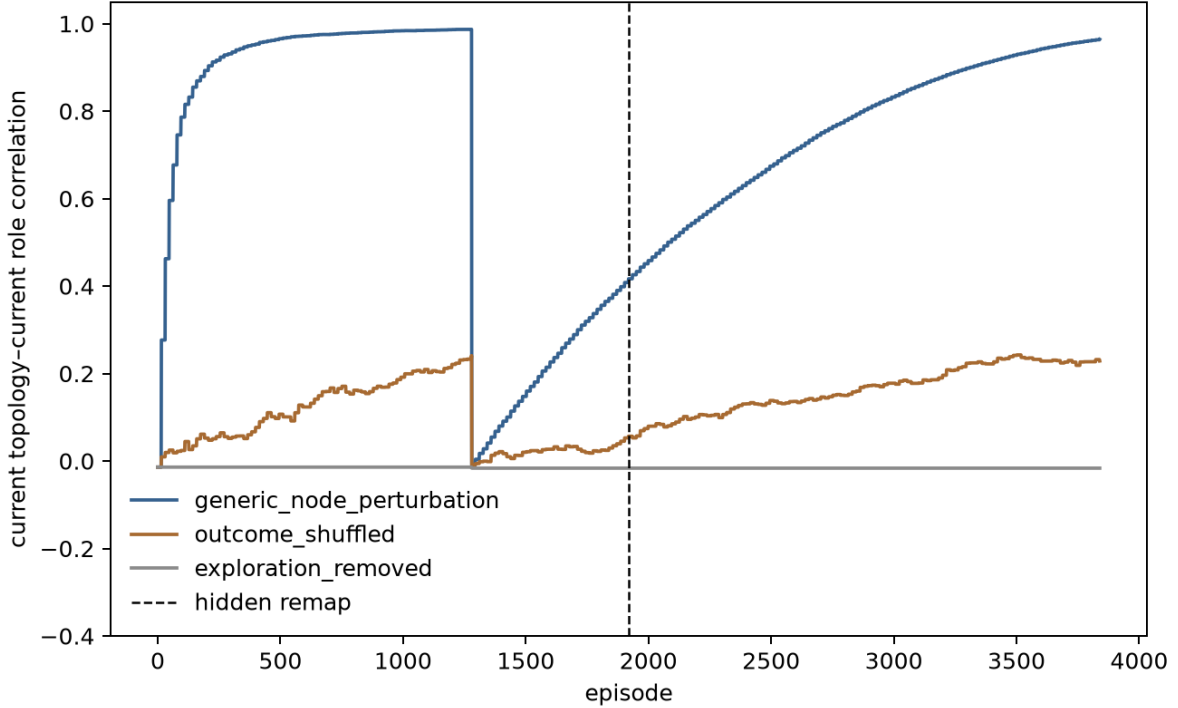


Figure 4: Frozen generic local-eligibility diagnostic at $N = 32$. The role alignment learned from scalar outcome and local perturbation eligibility rises during acquisition, drops when the hidden map changes, and is relearned. Shuffled outcome learns only weak alignment, while removing exploration eliminates learning.

biological or computational test should therefore preregister both the candidate causal variable and the observation model that maps it to the measured dendritic signal.

More broadly, the source audit prevents architectural resemblance from being mistaken for derivation. pART and SMART motivate a plausible route from selected representations to local plasticity. The exploratory eligibility rule supplies a computationally effective missing operation, but it is an added hypothesis. The most productive synthesis is therefore modular: published theory contributes representation selection, expectation, matching, and local plasticity motifs; cellular exploratory credit remains a distinct mechanism to test.

6 Limitations

The tasks are abstract simulations rather than reconstructions of retrosplenial circuitry, animal behavior, or complete pART and SMART equations. Binding a pART-selected representation to a SMART-like projection, targeting relevant BCI cells, mapping center/surround organization onto NDNF circuitry, and converting modeled activity into a calcium residual all require additional biological assumptions. The prestructured routing model is structurally favored, and the random-repertoire and outstar studies remain tied to their specified banks and ownership rules.

The integrated composition used sparse, quantized local plasticity. Its post-hoc analyses diagnose a failure but do not change the frozen classification. The local-eligibility comparator uses a known population-scaling correction; this carries no role information, yet success in a finite normalized task does not establish biological scalability. Finally, the manuscript is a synthesis of archived computational records, not an independent reproduction or peer review of either the simulations or the motivating animal experiment.

7 Biological predictions and discriminating tests

The computational alternatives make several separable predictions.

1. **Eligibility–outcome interaction.** Spontaneous or imposed cell-specific deviations before outcome should leave a local trace whose interaction with later scalar outcome predicts signed plasticity. Disrupting the trace–outcome interval should impair learning while preserving a scalar reward response.
2. **Within-cell remapping.** If selected feedback owns the expressed pattern, a context or contingency change should reorganize feedback before later somatic change and should permit the same cell to reverse sign across contexts.
3. **Center/surround measurement.** Simultaneous measurement and independent perturbation of excitatory feedback and NDNF-associated inhibitory surround should test whether a signed membrane influence, rather than an excitatory calcium proxy alone, carries the predictive signal.
4. **Projection- and phase-specific intervention.** Perturbing a candidate feedback source before outcome while preserving sensory error and reward should reveal whether that projection contributes cell-specific information. Broad activity suppression would be less diagnostic.

Together these tests could distinguish a routed-vector account, selection of an available representation, and locally computed exploratory credit.

8 Conclusion

The experiments show that scalar outcome can solve delayed representation credit without supplying a neuron-indexed teaching vector. In the tested pART-inspired compositions, however, cell-specific topology was chiefly inherited from available population patterns, and the preregistered dendrite-to-synapse-to-soma signature was not established. A generic three-factor rule learned arbitrary topology by combining the scalar outcome with cell-local exploratory eligibility. The distinction is the central result: scalar reward can choose among representations, but arbitrary cellular credit requires an additional cell-varying quantity. Determining whether dendrites carry that quantity, receive an explicit vector, or express matched feedback remains an empirical problem.

Data and code availability

All computational results reported here derive from archived protocols, frozen analyses, held-out experiment records, and explicitly labeled post-hoc audits accompanying this study. No new simulation results were generated specifically for the manuscript. The public project repository provides the source code, experimental records, analysis outputs, and figure files used here.

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